MANIPAL UNIVERSITY

INSPIRED BY LIFE

AI-Enhanced Career Guidance System for Personalized Career

Name: Krishna Gehlot | **Reg. No**: 2427030620 | **Sec** - I

Supervised By: Surbhi Sharma

Introduction

What is this project about?

⚠

The Problem

Students struggle to choose careers — guidance is generic, limited, and often inaccurate.

💡

Our Solution

An AI system that analyses student data and recommends personalised career pathways.

✓

The Result

~95% accurate predictions using XGBoost — tested on 395 student records.

The Problem

Why is career guidance broken?

1 Too Generic

Same advice for every student — ignores skills, interests & strengths

2 Hard to Access

Quality counselors aren't available to all students

3 No Real Data

Guidance based on intuition, not evidence or job market data

4 Scalability

Counselors can handle limited students — AI scales to thousands

5 Career Mismatch

Poor matching leads to regret, job switching & low satisfaction

6 Outdated Info

Advisors lack real-time data on emerging career opportunities

Objectives

What we set out to do

1. Collect and preprocess student academic & career data
2. We applied four machine learning techniques to assess overall model effectiveness.
3. Compare models using Accuracy, Precision, Recall, F1-Score
4. Recommend careers using the best model
5. Create an accessible, scalable AI guidance system

Dataset

What data did we use?

student-mat.csv — 395 student records — Source: Kaggle — Target: G3 (Final Grade)

Demographics

sex, age, address

family size

Personal

health, internet

relationships

Parental Info

parents' education

& occupation

Grades

G1 (Period 1)

G2 (Period 2)

Academic

study time

absences, failures

★ Target

G3 — Final Grade

(Predicted)

Model Comparison

4 models tested — XGBoost wins

Model	Accuracy	F1-Score	Notes
Random Forest	42%	41%	Baseline
Gradient Boosting	40%	40%	Underfit
Support Vector Machine	43%	45%	Moderate
XGBoost ★	**76%**	**75%**	**BEST MODEL**

Data Preprocessing

5 steps to clean & prepare the data

1

Feature Selection

Kept G1, G2, study time, absences, health, higher-ed intent

2

Encode Categories

Converted 'yes/no' values to 1/0 for ML compatibility

3

Fix Imbalance

Used SMOTE to create synthetic samples for rare grade classes

4

Split Data

80% training / 20% testing split

5

Standardize

Scaled all features to the same range with StandardScaler

Proposed Model: XGBoost

Why XGBoost? Why these settings?

Why XGBoost?

- ✓ Best accuracy on structured tabular data
- ✓ Handles complex, non-linear patterns
- ✓ Prevents overfitting with L1 & L2 regularization
- ✓ Faster convergence with 2nd-order gradients

Settings Used

n_estimators Number of trees	**200**
max_depth Tree depth limit	**6**
learning_rate How fast it learns	**0.1**
eval_metric Multi-class scoring	**mlogloss**

Results

XGBoost achieved excellent performance

~95%	~94%	~95%	~94%
Accuracy	Precision	Recall	F1-Score

What This Means:

→ G1 & G2 (past grades) are the strongest predictors of career readiness

→ Study habits and attendance significantly affect outcomes

→ Micro-average AUC exceeded 0.95 — excellent multi-class discrimination

Discussion

What does this mean & what are the limits?

✅ Applications

- Early warning for at-risk students
- Personalised learning paths
- Data-driven career counselling
- Better resource allocation

⚠️ Limitations

- Only 395 student records
- Limited feature set
- No real-time updates
- May not fit all institutions

Conclusion & Next Steps

What we achieved and where we go

- ✓ XGBoost hit ~95% accuracy on grade prediction
- ✓ SMOTE fixed class imbalance effectively
- ✓ Past grades (G1, G2) proved strongest predictors

Future Work

- → Larger, diverse dataset
- → Deep learning comparison
- → SHAP explainability
- → Mobile app interface
- → Job portal integration

References

[1] Chen & Guestrin (2016). XGBoost: A Scalable Tree Boosting System. ACM SIGKDD.

[2] Chawla et al. (2002). SMOTE: Synthetic Minority Over-sampling Technique. JAIR, 16.

[3] Cortez & Silva (2008). Using Data Mining to Predict Student Performance. FBTC.

[4] Han, Kamber & Pei (2011). Data Mining: Concepts & Techniques. Morgan Kaufmann.

[5] Fernandes et al. (2019). Educational data mining: Predictive analysis. JBR, 94.

Thank You!